

An Exploration on Modeling Haptic Reaction Time of 3D Interactive Tasks within Virtual Environments

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ABSTRACT

Virtual environments (VEs) utilize haptic cues – e.g., vibrotactile and force feedbacks – to facilitate user interactions in 3D tasks. Existing studies reported maximum likelihood estimation (MLE) to integrate cues of different modalities. The MLE integration was excluded for cues of the same modality. Although proportional likelihood estimation (PLE) was able to integrate same-modality cues for the parameter of task accuracy, its applicability remains unclear to other task parameters like reaction time. This feasibility study compared thus MLE and PLE to integrate the haptic cues for reaction time. PLE was found to be applicable to model haptic reaction time.

Keywords: virtual reality (VR); haptic cue; force; vibration.

Index Terms: #H.1.2 [User/Machine Systems]: Human information processing; H.5.1 [Multimedia Information Systems]: Artificial, augmented, and virtual realities; H.5.2 [User Interfaces]: Haptic I/O

1 INTRODUCTION

Haptic cues (e.g., feedbacks of vibration, force, texture, and temperature) are essential for user interaction with 3D objects in computer-based virtual environments (VEs) [1, 2]. Force and vibrotactile feedbacks are cues of particular interest due to hardware proliferation to render them. The mechanism underlying human cognition of these cues simultaneously (i.e., cue integration) remains underexplored although the cues can be delivered together to convey certain information. Existing studies reported that such an integration followed maximum likelihood estimation (MLE) for cues of different sensory modalities [3]. However, MLE failed to describe the integration of vibrotactile and force cues from the haptic modality [4]. Instead, the integration of these same-modality cues followed proportional likelihood estimation (PLE) for the parameter of task accuracy [4]. Since task accuracy is an accumulated outcome of several interactions, such accumulation limits PLE's use for real-time applications demanding the cue integration based on single interaction.

A candidate parameter without the accumulation would be reaction time – i.e., the interval between a cue onset and a user response. For this parameter, it is unknown however how MLE and PLE could be feasible to integrate vibrotactile and force cues based on single interaction. This study thus explored the feasibility of MLE and PLE to the haptic-cue integration for reaction time.

2 METHODS

Study Design: Sixteen healthy volunteers (22.7 ± 2.5 years old) participated in the study. Being naïve to the purpose of the study,

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all of them were right-handed, with normal or corrected-to-normal vision, and 3D stereo acuity of at least 40" arc.

A 3D stereoscopic VE was developed using Unity (2019.4.28f1). The visual scene of the VE, as shown in Fig. 1(a), consisted of a helical curve with its curvature and torsion being constant in space. Each participant performed an interactive task of tracing this curve. The participant used their right hand to hold the stylus of a haptic device (Omni; 3D Systems, USA) and to employ the stylus tip for the tracing. The stylus provided force cues in the direction of the curvature. A custom-built kit provided vibrotactile cues to the participant. The kit was constituted by a controller (Atmel SAM4S Xplained Pro; Atmel Corp., USA) and a pair of 10 mm vibrators (Precision Microdrives, UK). A Velcro band fastened the vibrators to the participant's right hand or forearm. The band was at the hand in the co-located (co) condition as depicted in Fig. 1(b) or at the forearm in the dis-located (dis) condition as shown in Fig. 1(c). An E4 bracelet (Empatica Inc., Italy) was worn on the left wrist of the participant, monitoring physiological signals for potential cybersickness.

The interactive task required the stylus tip to follow a partially transparent sphere, which moved along the curve at a constant speed of 2.0 cm/s. The initial point of the task was randomly placed on the curve for the participant to begin his/her tracing. After the tracing commenced, three force or vibrotactile cues were delivered randomly along the curve. The force cues had its magnitude of 500 mN with its direction tangential to the curve. The vibrotactile cues were at 200 HZ with an intensity of 0.9 G. All cues were 2.0 ~ 3.0 cm apart from each other. The force cues lasted 1.0 s long and the vibrotactile cues consisted of each vibrator active for 0.5 s consecutively. The participant was instructed to respond to a cue by pressing the buttons on the stylus as soon as possible. Twelve cues were delivered to the participant in a testing condition. Totally, there were six testing conditions as follows:

- V-co: only vibrotactile cue delivered at the hand
- V-dis: only vibrotactile cue rendered at the forearm
- F-co: only force cue provided to the hand with inactive vibrators at the hand
- F-dis: only force cue given to the hand with inactive vibrators at the forearm
- FV-co: both force and vibrotactile cues offered to the hand
- FV-dis: a force cue rendered at the hand and a vibrotactile cue delivered at the forearm

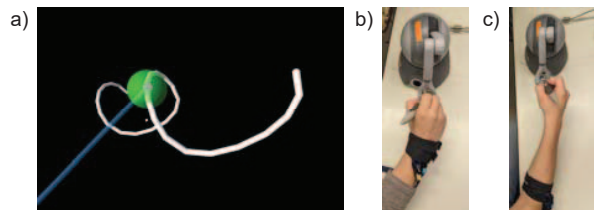


Figure 1: The setup of the VE: (a) its visual scene; (b) co-located cues; and (c) dis-located cues.

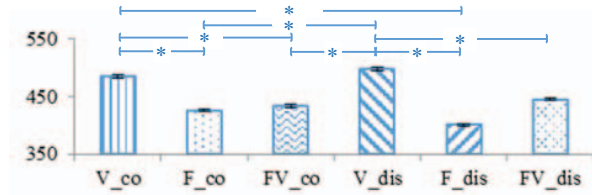


Figure 2: Reaction time (milliseconds along the vertical axis) of all testing conditions. [Error bars represent standard errors. The symbol of * denotes a Tukey's significance with $p < 0.05$.]

A training session preceded a session of the 6 testing conditions. After each testing condition, each participant answered a cybersickness and a NASA task load index (TLX) questionnaire [5]. The order of the conditions was counter-balanced among all participants. The participant spent about 1.0 hour in the study. The study had an ethics certificate approved by our institute.

Data Acquisition and Analyses: During the task, objective data were E4 signals and reaction time logged by the VE. Responses outside the duration of the cues were considered as incorrect to be removed. Subjective data were reported scores of cybersickness and cognitive workload gathered by the questionnaires. Both objective and subjective data were checked for their eligibility of ANOVA using the Kolmogorov-Smirnov normality test; and then were analyzed using one-way ANOVA and post-hoc Tukey's test.

3 RESULTS AND DISCUSSION

None of the participants suffered from cybersickness as confirmed by their cybersickness scores and physiological signals. Therefore, the data of all participants were used in the analyses.

ANOVA analyses of cognitive workload revealed no significant differences among all conditions [$F(5, 95) = 195.7, p > 0.05$]. This revelation agreed with similar observations of existing studies [4, 6], confirming the suitability of the data for MLE and PLE analyses.

As shown in Fig. 2, average reaction time of all conditions was as: 485.2 ± 51.4 ms (V-co), 426.3 ± 49.4 ms (F-co), 434.4 ± 49.4 ms (FV-co), 498.4 ± 61.3 ms (V-dis), 400.9 ± 32.1 ms (F-dis), and 445.7 ± 34.9 ms (FV-dis). The normality test revealed that the reaction time of each condition was normally distributed. ANOVA analyses of reaction time indicated a significant difference among all conditions [$F(5, 95) = 9.05, p < 0.05$]. Post-hoc analyses revealed significant differences between 7 pairs of the conditions: V-co vs. F-co, V-co vs. FV-co, V-co vs. F-dis, V-dis vs. F-co, V-dis vs. FV-co, V-dis vs. F-dis, and V-dis vs. FV-dis. Interestingly, the conditions with only vibrotactile cue yielded longer reaction time than other conditions. Compared to the force cues tangential to the curve, the vibrotactile cues might be unnatural for the motion of tracing the curve and thus delay the responses.

Additionally, MLE and PLE were applied to reaction time of all conditions. As shown in Fig. 3(a), the observed cue integration for the co-located condition, FV-co, did not follow MLE instead PLE. That is, the integration of two same-modality cues close to each other is non-optimal, because PLE was a special case of MLE to indicate a suboptimal integration. This suboptimality does not mean to ignore one and favor another [7]. This observation might result from the resource demand of the similar brain areas to process the cues simultaneously [8]. Consequently, the same-modality cues were interpreted as if there were only a single cue. In contrast, both MLE and PLE explained the observed cue integration for the dis-located condition, FV-dis, as seen in Fig. 3(b). In other words, the integration of two haptic cues separate sufficiently apart was optimal similar as that of different-modality cues [6]. All these observations found that PLE was feasible to explain the integration of same-modality cues co-located or dis-

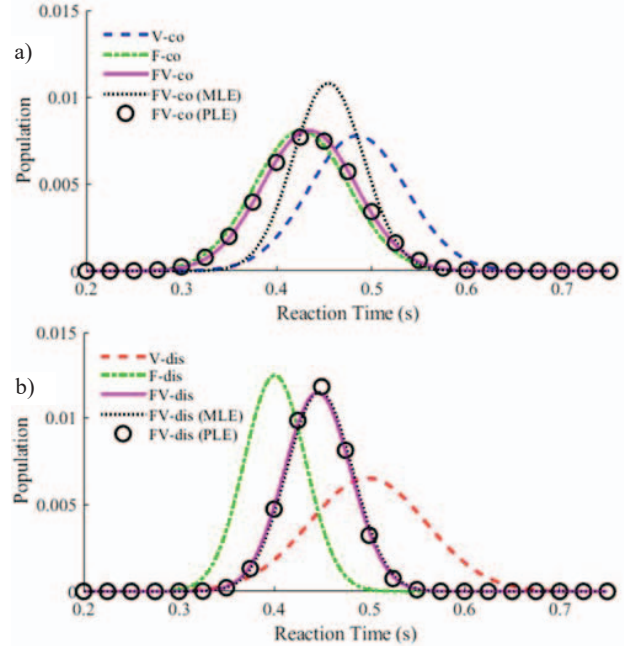


Figure 3: Gaussian-distributed estimations of MLE and PLE on reaction time in: (a) the co-located and (b) dis-located conditions.

located for reaction time. Further investigations are needed to examine the applicability of PLE to other task parameters.

4 CONCLUSION

This study revealed the feasibility of PLE to model haptic cue integration for reaction time of 3D interactive tasks within VEs.

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